

Weekly Progress Report – Week 03

Name: Somya Darshini Panda

Project Title: Exploratory Data Analysis (EDA) on Geldium Dataset

Week: 03

1. Work Completed During the Week

- Performed detailed data cleaning by handling missing values and removing duplicate records.
- Verified and corrected inconsistent data entries.
- Conducted Exploratory Data Analysis (EDA) using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
- Created histograms, bar charts, box plots, and correlation heatmaps.
- Analyzed relationships between variables and documented key observations.

2. Milestones Achieved

- Completed data cleaning and preprocessing.
- Generated meaningful visualizations.
- Identified trends, correlations, and outliers.
- Improved dataset quality for further analysis.

3. Challenges and Hurdles

Challenges included missing values, inconsistent records, outliers, and selecting suitable visualizations. These were addressed through data preprocessing, duplicate removal, handling missing values, and validating results before documentation.

4. Lessons Learned

- Understood the importance of data preprocessing.
- Improved skills in Python, Pandas, NumPy, Matplotlib, and Seaborn.
- Strengthened analytical thinking, visualization, and problem-solving skills.

Conclusion

Week 03 focused on exploratory analysis of the Geldium dataset. The activities enhanced my practical knowledge of data analytics and prepared me for the next phase of the project.